

Problem 1

(a) Express $P(X_i = 1)$ as a function of a and π .

We have

$$\begin{aligned} P(X_i = 1) &= P(X_i = 1, \text{Group} = A) + P(X_i = 1, \text{Group} = B) \\ &= P(X_i = 1 \mid \text{Group} = A) \cdot P(\text{Group} = A) + P(X_i = 1 \mid \text{Group} = B) \cdot P(\text{Group} = B) \\ &= a \cdot P(\text{Group} = A) + (1 - a) \cdot P(\text{Group} = B) \\ &= a \cdot \pi + (1 - a) \cdot (1 - \pi) . \end{aligned}$$

Similarly, we have

$$\begin{aligned} P(X_i = 0) &= P(X_i = 0, \text{Group} = A) + P(X_i = 0, \text{Group} = B) \\ &= P(X_i = 0 \mid \text{Group} = A) \cdot P(\text{Group} = A) + P(X_i = 0 \mid \text{Group} = B) \cdot P(\text{Group} = B) \\ &= (1 - a) \cdot P(\text{Group} = A) + a \cdot P(\text{Group} = B) \\ &= (1 - a) \cdot \pi + a \cdot (1 - \pi) . \end{aligned}$$

(Optional) Because the random variable X_i can only have 2 potential values, we can now easily check whether our expressions are correct as $P(X_i = 1) + P(X_i = 0) = 1$. We have

$$\begin{aligned} P(X_i = 1) + P(X_i = 0) &= a \cdot \pi + (1 - a) \cdot (1 - \pi) + (1 - a) \cdot \pi + a \cdot (1 - \pi) \\ &= a \cdot \pi + (1 - \pi) - a + \pi \cdot a + \pi - \pi \cdot a + a - \pi \cdot a \\ &= a \cdot \pi + 1 - \pi + \pi - \pi \cdot a \\ &= 1 . \end{aligned}$$

(b) Give expressions for the joint probability of the observed data $X_1, X_2, \dots, X_n$, and the corresponding likelihood function and loglikelihood function.

We aim to express the computation for

$$P(X_1, X_2, \dots, X_n) .$$

Because we know that the n samples are drawn with the i.i.d. assumption, it holds that

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i) .$$

Hence,

$$\begin{aligned}
 P(X_1, X_2, \dots, X_n) &= \prod_{i=1}^n P(X_i) \\
 &= \prod_{i=1}^n \begin{cases} P(X_i = 1) & \text{if } X_i = 1 \\ P(X_i = 0) & \text{if } X_i = 0 \end{cases} \\
 &= P(X_i = 1)^m \cdot P(X_i = 0)^{n-m} \\
 &= (a \cdot \pi + (1-a) \cdot (1-\pi))^m \cdot ((1-a) \cdot \pi + a \cdot (1-\pi))^{n-m} \tag{1}
 \end{aligned}$$

is the likelihood function of the observed samples.

Let us now derive the expression for the log likelihood of the data:

$$\begin{aligned}
 \log P(X_1, X_2, \dots, X_n) &= \log \left[(a \cdot \pi + (1-a) \cdot (1-\pi))^m \cdot ((1-a) \cdot \pi + a \cdot (1-\pi))^{n-m} \right] \\
 &= m \cdot \log [a \cdot \pi + (1-a) \cdot (1-\pi)] + (n-m) \cdot \log [(1-a) \cdot \pi + a \cdot (1-\pi)] \tag{2}
 \end{aligned}$$

(c) Derive an expression for the maximum likelihood estimator π_{ML} of π .

We now aim to estimate π . To this end, we denoted the estimated value for π as π_{ML} in Equation 2 and derive the log likelihood w.r.t. π_{ML} . We have

$$\begin{aligned}
 &\frac{\partial}{\partial \pi_{ML}} \log P(X_1, X_2, \dots, X_n) \\
 &= \frac{\partial}{\partial \pi_{ML}} m \cdot \log [a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})] + (n-m) \cdot \log [(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})] \\
 &= m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) + (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a) - a) .
 \end{aligned}$$

We now find the critical points by setting the gradient to zero:

$$\begin{aligned}
 &\frac{\partial}{\partial \pi_{ML}} \log P(X_1, X_2, \dots, X_n) \stackrel{!}{=} 0 \\
 m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) + (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a) - a) &= 0 \tag{3}
 \end{aligned}$$

where Equation 3 holds whenever $(1-a) = a$ as both additive terms are zero. It is easy to see that $a = 0.5$ is the only real solution for this equation. However, as we are interested in the maximum likelihood estimator of π_{ML} , setting $a = 0.5$ prohibits us from finding the maximum value. This is because if $a = 0.5$, we have $P(X_i = 1) = 0.5 = P(X_i = 0)$, and therefore the likelihood function is constant, regardless of the value of π . Therefore, we need to consider the case where $a \neq 0.5$.

We now solve Equation 3 for π_{ML} while assuming that $q \neq 0.5$. We have:

$$\begin{aligned}
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) &= -(n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a) - a) \\
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) &= (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot (a - (1-a)) \\
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} &= (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \\
m \cdot \frac{1}{(2a-1) \cdot \pi_{ML} + 1-a} &= (n-m) \cdot \frac{1}{(1-2a) \cdot \pi_{ML} + a} \\
m \cdot \frac{1}{(2a-1) \cdot \pi_{ML} + 1-a} &= (n-m) \cdot \frac{1}{a - (2a-1) \cdot \pi_{ML}} \\
m \cdot (a - (2a-1) \cdot \pi_{ML}) &= (n-m) \cdot ((2a-1) \cdot \pi_{ML} + 1-a) \\
am - (2a-1) \cdot m \cdot \pi_{ML} &= (n-m) \cdot (2a-1) \cdot \pi_{ML} + (n-m) \cdot (1-a) \\
am - (n-m) \cdot (1-a) &= (n-m) \cdot (2a-1) \cdot \pi_{ML} + (2a-1) \cdot m \cdot \pi_{ML} \\
am - (n-m) \cdot (1-a) &= \pi_{ML} \cdot (2a-1)((n-m) + m) \\
am - (n-m) + a \cdot n - am &= \pi_{ML} \cdot (2a-1)(n) \\
n \cdot (a-1) + m &= \pi_{ML} \cdot (2a-1)(n) \\
\frac{n \cdot (a-1) + m}{(2a-1)(n)} &= \pi_{ML} \\
\frac{n \cdot (a-1) + m}{n} \cdot \frac{1}{2a-1} &= \pi_{ML} \\
\left(a-1 + \frac{m}{n}\right) \cdot \frac{1}{2a-1} &= \pi_{ML} \tag{4}
\end{aligned}$$

We verify that this is indeed a maximum by plugging the result from Equation 4 into the second derivative.

For ease of notation, we will use $b = (1-a)$. We have

$$\begin{aligned}
&\frac{\partial^2}{\partial \pi_{ML}^2} \log P(X_1, X_2, \dots, X_n) \\
&= \frac{\partial}{\partial \pi_{ML}} m \cdot (a - (1-a)) \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} + (n-m) \cdot ((1-a) - a) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \\
&= -m \cdot (a - (1-a)) \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} \cdot (a - (1-a)) \\
&\quad - (n-m) \cdot ((1-a) - a) \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \cdot ((1-a) - a) \\
&= -m \cdot (a - (1-a)) \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} \cdot (a - (1-a)) \\
&\quad - (n-m) \cdot (a - (1-a)) \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \cdot (a - (1-a)) \\
&= -m \cdot (a - (1-a))^2 \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} - (n-m) \cdot (a - (1-a))^2 \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \\
&= -m \cdot (a-b)^2 \cdot \frac{1}{(a \cdot \pi_{ML} + b \cdot (1-\pi_{ML}))^2} - (n-m) \cdot (a-b)^2 \cdot \frac{1}{(b \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \\
&\leq -m \cdot \frac{1}{(a \cdot \pi_{ML} + b \cdot (1-\pi_{ML}))^2} - (n-m) \cdot \frac{1}{(b \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2}
\end{aligned}$$

$$\begin{aligned}
&= -m \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} - (n-m) \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \\
&= -m \cdot \alpha - (n-m) \cdot \beta \\
&\leq -m - (n-m) \\
&\leq 0
\end{aligned}$$

with $\alpha > 0$ and $\beta > 0$ from the squaring of the denominator. Hence, we know that the second derivative is negative whenever $n > 0$. From this, it follows that our estimator π_{ML} is indeed a maximum.

(d) *Can you think of an easier way to arrive at the same estimator as the maximum likelihood estimator?*

We observe that X_i follows a Bernoulli distribution. We now first show that this holds.

Recall that the definition of a Bernoulli distribution is given by

$$\begin{aligned}
P(X = 1) &= p \\
P(X = 0) &= 1 - p .
\end{aligned}$$

Therefore, we now aim to show that $P(X_i)$ follows a Bernoulli distribution. From (a), we have that We have

$$P(X_i = 1) = a \cdot \pi + (1-a) \cdot (1-\pi)$$

and

$$P(X_i = 0) = (1-a) \cdot \pi + a \cdot (1-\pi) .$$

We multiply out the expression to get

$$\begin{aligned}
P(X_i = 1) &= a \cdot \pi + 1 - \pi - a \cdot (1-\pi) \\
&= a \cdot \pi + 1 - \pi - a + a \cdot \pi \\
&= 2a \cdot \pi + 1 - \pi - a \\
&= 1 + 2a \cdot \pi - \pi - a
\end{aligned} \tag{5}$$

$$\begin{aligned}
P(X_i = 0) &= (1-a) \cdot \pi + a \cdot (1-\pi) \\
&= \pi - a \cdot \pi + a - a \cdot \pi \\
&= -2a \cdot \pi + \pi + a .
\end{aligned} \tag{6}$$

Therefore, with $p = 1 + 2a \cdot \pi - \pi - a$ (from Equation 5), we have

$$\begin{aligned}
 P(X = 0) &= 1 - p \\
 &= 1 - P(X = 1) \\
 &= 1 - (1 + 2a \cdot \pi - \pi - a) \\
 &= 1 - 1 - 2a \cdot \pi + \pi + a \\
 &= -2a \cdot \pi + \pi + a
 \end{aligned} \tag{7}$$

where Equation 7 matches the expression in Equation 6. Therefore, we have shown that X_i follows a Bernoulli distribution with parameter $p = 1 + 2a \cdot \pi - \pi - a$.

Following from this result, because the random variable m is the sum of n i.i.d. Bernoulli random variables, we know that m follows a Binomial distribution with parameters n and p . Hence, the likelihood of observing m ones in n samples is given by the probability density function of the Binomial distribution:

$$\Pr(X = m) = \binom{n}{m} p^m (1 - p)^{n-m} . \tag{8}$$

While Equation 8 differs from the likelihood function we derived in Equation 1, we can see that the $\binom{n}{m}$ term cancels out when deriving the log likelihood; First, the log likelihood transforms the multiplication into a sum with a log term, and then when deriving w.r.t. π_{ML} , the $\log \binom{n}{m}$ becomes zero and therefore disappears. Hence, we end up with the same expression for the log likelihood.

As the Binomial distribution is well-known, we know that the maximum likelihood estimator of p of the Binomial distribution is given by $p_{ML} = \frac{m}{n}$, which is the proportion of successes, i.e., the number of $X_i = 1$ in n samples.¹

Finally, to derive the maximum likelihood estimator of π_{ML} , we equate p_{ML} and $P(X_i = 1)$ (as $P(X_i = 1) = p$), and then solve for π_{ML} :

$$\begin{aligned}
 P(X_i = 1) &= p_{ML} \\
 1 + 2a \cdot \pi_{ML} - \pi_{ML} - a &= \frac{m}{n} \\
 2a \cdot \pi_{ML} - \pi_{ML} &= \frac{m}{n} + a - 1 \\
 \pi_{ML} \cdot (2a - 1) &= \frac{m}{n} + a - 1 \\
 \pi_{ML} &= \frac{\frac{m}{n} + a - 1}{(2a - 1)} .
 \end{aligned} \tag{9}$$

We clearly observe that Equation 9 matches Equation 4, hence, we found the same maximum likelihood estimator of π_{ML} as in (c).

(e) Show that π_{ML} is an unbiased estimator of π .

We aim to show that $\mathbb{E}[\pi_{ML}] = \pi$. To this end, we will make use of the knowledge introduced in (d), namely

¹See https://en.wikipedia.org/wiki/Binomial_distribution#Estimation_of_parameters.

that X_i follows a Bernoulli distribution.

We have

$$\begin{aligned}\mathbb{E}[\pi_{ML}] &= \mathbb{E}\left[\frac{\frac{m}{n} + a - 1}{(2a - 1)}\right] \\ &= \mathbb{E}\left[\frac{\frac{\sum_{i=1}^n X_i}{n} + a - 1}{(2a - 1)}\right] \\ &= \frac{\sum_{i=1}^n \mathbb{E}[X_i] + a \cdot n - 1 \cdot n}{(2a - 1) \cdot n}.\end{aligned}\tag{10}$$

We now compute the expectation of X_i :

$$\begin{aligned}\mathbb{E}[X_i] &= P(X_i = 1) \cdot 1 + P(X_i = 0) \cdot 0 \\ &= P(X_i = 1) \\ &= a \cdot \pi + (1 - a) \cdot (1 - \pi) \\ &= (1 - a) + (2a - 1) \cdot \pi.\end{aligned}\tag{11}$$

Finally, we can substitute Equation 11 into Equation 10 to get

$$\begin{aligned}\mathbb{E}[\pi_{ML}] &= \frac{\sum_{i=1}^n \mathbb{E}[X_i] + a \cdot n - 1 \cdot n}{(2a - 1) \cdot n} \\ &= \frac{\sum_{i=1}^n ((1 - a) + (2a - 1) \cdot \pi) + a \cdot n - 1 \cdot n}{(2a - 1) \cdot n} \\ &= \frac{n \cdot ((1 - a) + (2a - 1) \cdot \pi) + a \cdot n - 1 \cdot n}{(2a - 1) \cdot n} \\ &= \frac{((1 - a) + (2a - 1) \cdot \pi) + a - 1}{(2a - 1)} \\ &= \frac{-(a - 1) + (2a - 1) \cdot \pi + (a - 1)}{(2a - 1)} \\ &= \frac{(2a - 1) \cdot \pi}{(2a - 1)} \\ &= \pi.\end{aligned}\tag{12}$$

We have shown that the expected value of π_{ML} is equal to the true value of π (Equation 12). Therefore, we can conclude that our estimator for π_{ML} is unbiased.

(f) *Derive an expression for the variance of π_{ML} . Analyse its dependence on a .*

We have

$$\begin{aligned}
\mathbb{V}[\pi_{ML}] &= \mathbb{V}\left[\frac{\frac{m}{n} + a - 1}{(2a - 1)}\right] \\
&= \mathbb{V}\left[\frac{m}{n} + a - 1\right] \cdot \frac{1}{(2a - 1)^2} \\
&= \mathbb{V}\left[\frac{1}{n}(m + a \cdot n - 1 \cdot n)\right] \cdot \frac{1}{(2a - 1)^2} \\
&= \mathbb{V}[m + a \cdot n - 1 \cdot n] \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= \mathbb{V}\left[\sum_{i=1}^n X_i + a \cdot n - 1 \cdot n\right] \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= \mathbb{V}\left[\sum_{i=1}^n X_i\right] \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= \sum_{i=1}^n \mathbb{V}[X_i] \cdot \frac{1}{(2a - 1)^2 \cdot n^2} .
\end{aligned} \tag{13}$$

We now compute the variance of X_i . We have:

$$\begin{aligned}
\mathbb{E}[X_i^2] &= P(X_i = 1) \cdot 1^2 + P(X_i = 0) \cdot 0^2 \\
&= P(X_i = 1) \\
&= (1 - a) + (2a - 1) \cdot \pi ,
\end{aligned} \tag{14}$$

and

$$\begin{aligned}
\mathbb{E}[X_i]^2 &= ((1 - a) + (2a - 1) \cdot \pi)^2 \\
&= (1 - a)^2 + (2a - 1)^2 \cdot \pi^2 + 2 \cdot (1 - a) \cdot (2a - 1) \cdot \pi \\
&= 1^2 + a^2 - 2a + 4a^2 \pi^2 + \pi^2 - 4a \cdot \pi^2 + (2 - 2a) \cdot (2a - 1) \cdot \pi \\
&= 1 + a^2 - 2a + 4a^2 \pi^2 + \pi^2 - 4a \cdot \pi^2 + (4a - 2 - 4a^2 + 2a) \cdot \pi \\
&= 1 + \pi (6a - 4a^2 - 2) + \pi^2 (4a^2 - 4a + 1) - 2a + a^2 .
\end{aligned} \tag{15}$$

Therefore, the variance of X_i is given by

$$\begin{aligned}
V(X_i) &= \mathbb{E}[X_i^2] - E[X_i]^2 \\
&= \pi(2a - 1) + 1 - a - (1 + \pi(6a - 4a^2 - 2) + \pi^2(4a^2 - 4a + 1) - 2a + a^2) \\
&= \pi(2a - 1) + 1 - a - 1 - \pi(6a - 4a^2 - 2) - \pi^2(4a^2 - 4a + 1) + 2a - a^2 \\
&= \pi(2a - 1) - \pi(6a - 4a^2 - 2) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi(2a - 1 - (6a - 4a^2 - 2)) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi(2a - 1 - 6a + 4a^2 + 2) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi(1 - 4a + 4a^2) - \pi^2(1 - 4a + 4a^2) + a - a^2 \\
&= (1 - 4a + 4a^2) \cdot (\pi - \pi^2) + a - a^2.
\end{aligned} \tag{16}$$

Finally, to compute the variance of the estimator of π_{ML} , we plug Equation 16 into Equation 13 and get

$$\begin{aligned}
\mathbb{V}[\pi_{ML}] &= \sum_{i=1}^n \mathbb{V}[X_i] \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= \sum_{i=1}^n ((1 - 4a + 4a^2) \cdot (\pi - \pi^2) + a - a^2) \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= n \cdot ((1 - 4a + 4a^2) \cdot (\pi - \pi^2) + a - a^2) \cdot \frac{1}{(2a - 1)^2 \cdot n^2} \\
&= ((1 - 4a + 4a^2) \cdot (\pi - \pi^2) + a - a^2) \cdot \frac{1}{(2a - 1)^2 \cdot n} \\
&= ((2a - 1)^2 \cdot (\pi - \pi^2) + a - a^2) \cdot \frac{1}{(2a - 1)^2 \cdot n} \\
&= \frac{(2a - 1)^2 \cdot (\pi - \pi^2)}{(2a - 1)^2 \cdot n} + \frac{a - a^2}{(2a - 1)^2 \cdot n} \\
&= \frac{\pi - \pi^2}{n} + \frac{a - a^2}{(2a - 1)^2 \cdot n}.
\end{aligned} \tag{17}$$

We correctly observe that the variance is undefined for $a = 0.5$. Furthermore, we observe that the variance decreases as n increases, which is to be expected as we get more evidence with more data. Lastly, we observe that whenever a approaches either 0 or 1, the variance decreases. The same phenomenon is observed for the true value of π .

Figure 1 shows the log-scaled variance of π_{ML} as a function of a and π , which clearly displays the behavior previously described. In short, our monte-carlo simulation confirms that the variance of the estimator of π_{ML} is maximal whenever a approaches 0.5, and furthermore increased as π approaches 0.5. This makes sense, as whenever $a = 0.5$ (or close), there is very little information gained from the answer of the users, a yes could be from both groups. Hence, the variance of the estimator will be very large in this case. Furthermore, whenever one group is much larger than the other (i.e., $\pi \rightarrow 0$ or $\pi \rightarrow 1$), we expect the answers to be more consistent, and hence the estimation of π will be more accurate. Thus, the variance decreases as π approaches the extremes of 0 and 1.

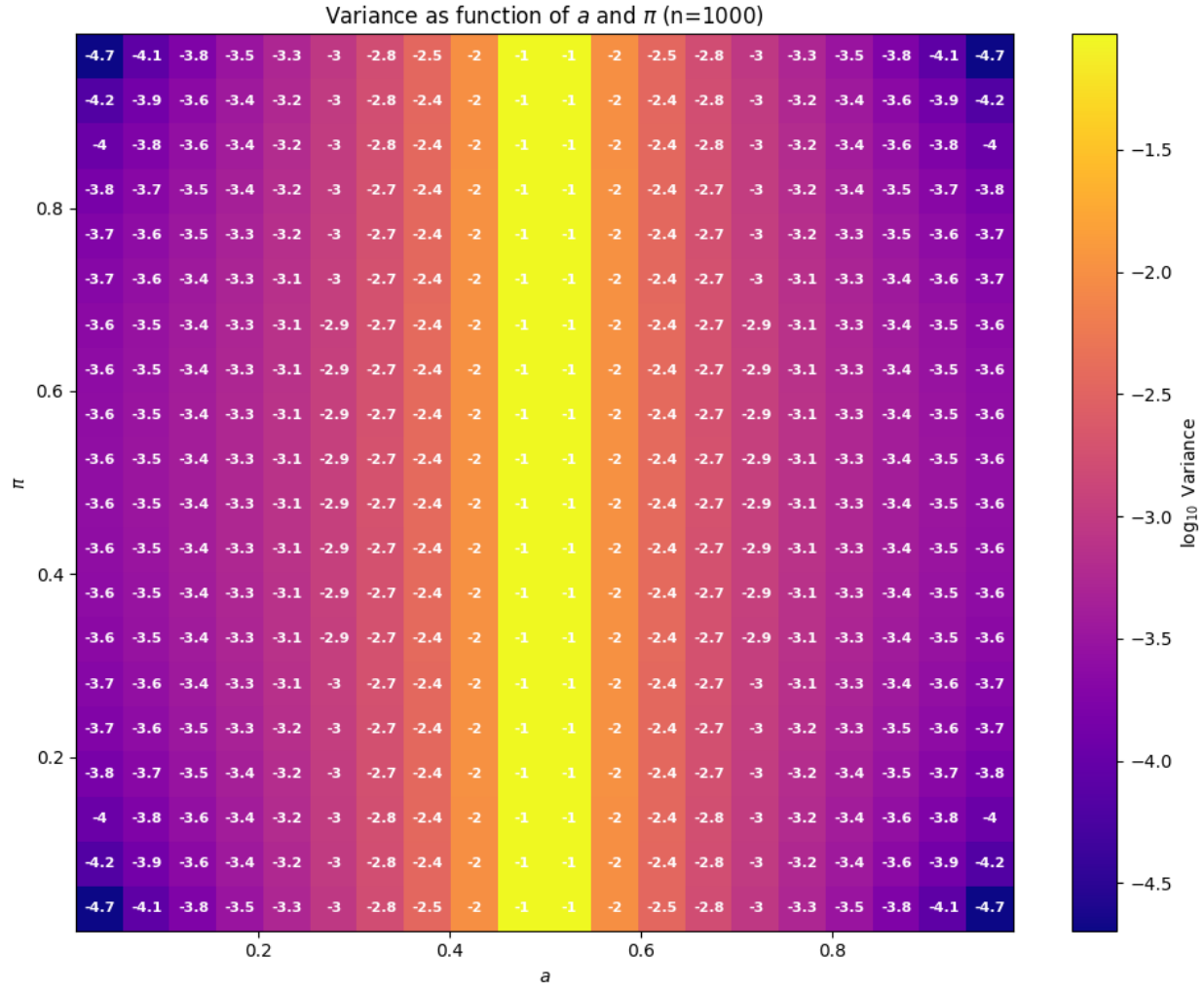


Figure 1: Variance of π_{ML} as function of a and π for different values of n . Here, the values are log-scaled for better display. Note that we restrict $a \in [0.01, 0.99]$ and $\pi \in [0.01, 0.99]$ for the plot. Hence, very low values are expected whenever a or π exceeds these bounds.

(g) Give an expression for the bias of this naive estimator. Is it asymptotically unbiased?

We aim to recover the value of π from direct questioning. However, persons belonging to the sensitive group will lie with probability ℓ , whereas persons belonging to the non-sensitive group will answer truthfully and therefore lie with probability 0.

Naively, we could estimate the value of π as the fraction of persons answering yes to the question:

$$\pi_{\text{naive}} = \frac{\sum_{i=1}^n X_i}{n}.$$

We can model the whether a person belongs to the sensitive group as a hidden variable $S_i \in \{0, 1\}$ with $S_i = 1$ describing that the person belongs to the sensitive group, and $S_i = 0$ describes that a person belongs to the non-sensitive group.

Naturally, we have that

$$\begin{aligned} P(S_i = 1) &= \pi \\ P(S_i = 0) &= 1 - \pi \end{aligned}$$

which is distributed according to a Bernoulli distribution with parameter $p = \pi$.

We can now express the probability of a person answering yes or no to the question by marginalizing over the hidden variable S_i :

$$\begin{aligned} P(X_i = 1) &= P(X_i = 1, S_i = 1) + P(X_i = 1, S_i = 0) \\ &= \pi \cdot (1 - \ell) + 0 \\ P(X_i = 0) &= P(X_i = 0, S_i = 1) + P(X_i = 0, S_i = 0) \\ &= \pi \cdot \ell + (1 - \pi) . \end{aligned}$$

We verify that the probabilities are correct by adding both $P(X_i = 1)$ and $P(X_i = 0)$ and getting 1:

$$\begin{aligned} P(X_i = 1) + P(X_i = 0) &= \pi \cdot (1 - \ell) + \pi \cdot \ell + (1 - \pi) \\ &= \pi(1 - \ell + \ell) + 1 - \pi \\ &= \pi(1) + 1 - \pi \\ &= 1 . \end{aligned}$$

We now compute the expected value of the naive estimator of π to check whether the estimator is unbiased. We have

$$\begin{aligned} \mathbb{E}[\pi_{\text{naive}}] &= \mathbb{E} \left[\frac{\sum_{i=1}^n X_i}{n} \right] \\ &= \frac{1}{n} \cdot \sum_{i=1}^n \mathbb{E}[X_i] \\ &= \frac{1}{n} \cdot \sum_{i=1}^n (P(X_i = 1) \cdot 1 + P(X_i = 0) \cdot 0) \\ &= \frac{1}{n} \cdot \sum_{i=1}^n (P(X_i = 1)) \\ &= \frac{1}{n} \cdot \sum_{i=1}^n (\pi \cdot (1 - \ell)) \\ &= \frac{1}{n} \cdot n (\pi \cdot (1 - \ell)) \\ &= \pi \cdot (1 - \ell) . \end{aligned} \tag{18}$$

We can now use the expected value of π_{naive} to compute the bias of the estimator:

$$\begin{aligned}
 \text{Bias}(\pi_{\text{naive}}, \pi) &= [\mathbb{E}[\pi_{\text{naive}}] - \pi]^2 \\
 &= [\pi \cdot (1 - \ell) - \pi]^2 \\
 &= [\pi - \pi\ell - \pi]^2 \\
 &= [-\pi\ell]^2 \\
 &= \pi^2 \ell^2.
 \end{aligned} \tag{19}$$

Therefore, only whenever $\ell = 0$, i.e., no one lies, is the naive estimator of π unbiased. Otherwise, the estimator is biased towards and goes towards zero bias whenever either ℓ or π goes to zero.

(h) *Derive expressions for the mean squared error (MSE) of the randomized response estimator and the naive estimator. Analyse the expressions to draw conclusions about when, depending on the relevant parameters (π, a, ℓ, n) , one estimator should be favored over the other.*

We aim to compute the mean squared error of our two estimators of π .

We use the Bias Variance decomposition to decompose the MSE into the bias and variance terms.² Therefore, we simplify the MSE into

$$\begin{aligned}
 \text{MSE}(\hat{\pi}, \pi) &= \mathbb{E} \left[(\hat{\pi} - \mathbb{E}[\hat{\pi}])^2 \right] + [\mathbb{E}[\hat{\pi}] - \pi]^2 \\
 &= \mathbb{V}[\hat{\pi}] + [\mathbb{E}[\hat{\pi}] - \pi]^2 \\
 &= \mathbb{V}[\hat{\pi}] + \text{Bias}(\hat{\pi}, \pi)
 \end{aligned}$$

Therefore, we have

²See https://en.wikipedia.org/wiki/Mean_squared_error#Proof_of_variance_and_bias_relationship.

$$\begin{aligned}
\mathbb{V}[\pi_{\text{naive}}] &= \mathbb{V}\left[\frac{\sum_{i=1}^n X_i}{n}\right] \\
&= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}[X_i] \\
&= \frac{1}{n^2} \sum_{i=1}^n (\mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2) \\
&= \frac{1}{n^2} \sum_{i=1}^n (P(X_i = 1) - (P(X_i = 1))^2) \\
&= \frac{1}{n^2} \sum_{i=1}^n (\pi \cdot (1 - \ell) - (\pi \cdot (1 - \ell))^2) \\
&= \frac{n}{n^2} \left((\pi \cdot (1 - \ell)) - (\pi \cdot (1 - \ell))^2 \right) \\
&= \frac{\pi \cdot (1 - \ell) - \pi^2 \cdot (1 - \ell)^2}{n} \tag{20}
\end{aligned}$$

$$\begin{aligned}
\text{MSE}(\pi_{\text{naive}}, \pi) &= \mathbb{V}[\pi_{\text{naive}}] + \text{Bias}(\hat{\pi}, \pi) \\
&\stackrel{(20)}{=} \frac{\pi \cdot (1 - \ell) - \pi^2 \cdot (1 - \ell)^2}{n} + \text{Bias}(\hat{\pi}, \pi) \\
&\stackrel{(19)}{=} \frac{\pi \cdot (1 - \ell) - \pi^2 \cdot (1 - \ell)^2}{n} + \pi^2 \ell^2 .
\end{aligned}$$

Because the π_{ML} is an unbiased estimator of π , we have that $\text{Bias}(\pi_{ML}, \pi) = 0$. Therefore, we have

$$\begin{aligned}
\text{MSE}(\pi_{ML}, \pi) &= \mathbb{V}[\pi_{ML}] + \text{Bias}(\pi_{ML}, \pi) \\
&= \mathbb{V}[\pi_{ML}] \\
&\stackrel{(17)}{=} \frac{\pi - \pi^2}{n} + \frac{a - a^2}{(2a - 1)^2 \cdot n} .
\end{aligned}$$

First, we observe that whenever no-one lies ($\ell = 0$), the left side of the MSE of π_{naive} is equal to the left side of the MSE of π_{ML} . However, as the right side of the MSE of π_{naive} is multiplied with ℓ^2 , the MSE is zero for $\ell = 0$, whereas the MSE of π_{ML} is non-zero, given $a \notin \{0, 1\}$. If $a \in \{0, 1\}$, then both estimators are equal w.r.t. the MSE.

Due to the complex expressions, finding the cut-off points for when both MSE expressions are equal — and therefore finding whenever one estimator is better than the other analytically — is very complicated. While we have explained the trivial edge cases, whenever we have $a \notin \{0, 1\} \wedge \pi \notin \{0, 1\}$, we have no easy way to determine when which estimator is better than the other. To overcome this issue, we have computed the expected MSE for both estimators for a range of values for a, π, ℓ , and n and show the results in Figure 2. Figure 2 clearly shows the complex, non-linear relationship between both MSE expression.

First, it is clear that no single best estimator exists for all parameters. The shape of the region where π_{naive} has a lower MSE than π_{ML} is quite clear: whenever $\ell = 0$, the naive estimator is better than the ML estimator. Furthermore, whenever $\pi = 0$, the naive estimator is also better. Both cases make sense, as whenever no one lies, the naive estimator is the better estimator of the two.

Figure 2 also shows that whenever a is close or equal to 0.5, the naive estimator is the better solution. Note that while our plot shows that in the subplot (d), for some parameters which include $a = 0.5$, the ML estimator is to be favored. This is not true, as it is simply an artefact of the plotting; We use only 100 points for each axis, and therefore the voxels are quite large. With increased resolution, the same plot would show that the naive estimator is the better solution for these parameters.

Lastly, for all other cases that we did not mention, the ML estimator is the better solution for large n . Refer to Figure 2 for the exact shape. This is logical, as the MSE of the naive estimator has a part independent of n , whereas the MSE of the ML estimator is entirely divided by n . Thus, for large n , the ML estimator will go towards zero, whereas the naive estimator will remain constant.

Problem 2

We have implemented the given formulas for estimating the population size $\hat{N}$ and its variance $\hat{\sigma}_{\hat{N}}$ given some probability distributions from which we sample the probability of catching one animal. We have run the Monte-Carlo simulations for 1000 runs for each given probability distribution and report on the computed values in Table 1 for $p_i \sim U(0, 0.01)$, Table 2 for $p_i \sim U(0, 0.02)$, Table 7 for $p_i \sim 0.02 \cdot \text{Beta}(1, 3)$, and Table 8 for $p_i \sim 0.01 \cdot \text{Beta}(2, 2)$.

Note that for the sake of ease of comparison, we report on $\hat{\sigma}_{\hat{N}} = \sqrt{\hat{\sigma}_{\hat{N}}^2}$ instead of the variance $\hat{\sigma}_{\hat{N}}^2$ as it allows us to directly compare it with the reported standard deviations of the other parameters.

In this question, we are interested in the differences that arise from the different probability distributions, besides their performances.

We first plot the probability distribution for the four probability distributions that we are investigating in Figure 3. From that figure, we hypothesize that the expected value of catching one animal is not the same for all four probability distributions. Therefore, let us first compute the expected value of catching one animal for each probability distribution.

For the uniform distribution $U(0, a)$, with $a \in \mathbb{R} \setminus \{0\}$, we have:

$$\begin{aligned}
 \mathbb{E}_{p \sim U(0, a)}[X_i^t] &= \mathbb{E}_{p \sim U(0, a), q \sim U(0, 1)}[q \leq p] \\
 &= \mathbb{E}_{p \sim U(0, a)}[\mathbb{E}_{q \sim U(0, 1)}[q \leq p]] \\
 &= \int_0^a f_p(p) \cdot \mathbb{E}_{q \sim U(0, 1)}[q \leq p] \, dp \\
 &= \int_0^a \frac{1}{a} \cdot \mathbb{E}_{q \sim U(0, 1)}[q \leq p] \, dp \\
 &= \frac{1}{a} \int_0^a \mathbb{E}_{q \sim U(0, 1)}[q \leq p] \, dp \\
 &= \frac{1}{a} \int_0^a \int_0^1 f_q(q) \cdot \mathbb{1}\{q \leq p\} \, dq \, dp \\
 &= \frac{1}{a} \int_0^a \int_0^1 1 \cdot \mathbb{1}\{q \leq p\} \, dq \, dp \\
 &= \frac{1}{a} \int_0^a \int_0^p 1 \, dq \, dp \\
 &= \frac{1}{a} \int_0^a [q]_0^p \, dp \\
 &= \frac{1}{a} \int_0^a p \, dp \\
 &= \frac{1}{a} \left[\frac{p^2}{2} \right]_0^a \\
 &= \frac{1}{a} \frac{a^2}{2} \\
 &= \frac{a}{2} .
 \end{aligned}$$

It naturally follows that

$$\mathbb{E}_{p \sim U(0,0.01)}[X_i^t] = 0.005 \quad (21)$$

and

$$\mathbb{E}_{p \sim U(0,0.02)}[X_i^t] = 0.01 . \quad (22)$$

For $p \sim \text{Beta}(\cdot)$, we have:

$$\begin{aligned} \mathbb{E}_{p \sim \text{Beta}(\alpha, \beta)}[X_i^t] &= \mathbb{E}_{p \sim \text{Beta}(\alpha, \beta), q \sim U(0,1)}[q \leq p] \\ &= \mathbb{E}_{p \sim \text{Beta}(\alpha, \beta)}[\mathbb{E}_{q \sim U(0,1)}[q \leq p]] \\ &= \int_0^1 f_{\text{Beta}(\alpha, \beta)}(p) \cdot \mathbb{E}_{q \sim U(0,1)}[q \leq p] \, dp \\ &= \int_0^1 f_{\text{Beta}(\alpha, \beta)}(p) \cdot \int_0^1 f_q(q) \cdot \mathbb{1}\{q \leq p\} \, dq \, dp \\ &= \int_0^1 f_{\text{Beta}(\alpha, \beta)}(p) \cdot p \, dp \\ &= \mathbb{E}_{X \sim \text{Beta}(\alpha, \beta)}[X] \\ &= \frac{\alpha}{\alpha + \beta} . \end{aligned} \quad (23)$$

Note that if $p \sim a \cdot \text{Beta}(\cdot)$ with $a \in \mathbb{R} \setminus \{0\}$, we can just multiply Equation 23 with a due to the properties of the expectation and the integrals. a would be contained in the density function as a multiplicative factor, and thus, we can extract it. Note that I got the expectation of the Beta distribution from wikipedia.³

Therefore, it follows that

$$\mathbb{E}_{p \sim 0.02 \cdot \text{Beta}(1,3)}[X_i^t] = 0.02 \cdot \frac{1}{4} = 0.005 \quad (24)$$

and

$$\mathbb{E}_{p \sim 0.01 \cdot \text{Beta}(2,2)}[X_i^t] = 0.01 \cdot \frac{1}{2} = 0.005 . \quad (25)$$

The usefulness of Equation 21–22 and Equation 24–25 expected values will become clear in the next section.

Expected Behavior. Before discussing the results of the Monte-Carlo simulations in detail, let us first investigate the expected behavior of the simulations.

We have computed the expected number of times an animal is being caught at each time step. Recall that the expected values for X_i^t is 0.005 for all given distributions, except when $p \sim U(0,0.01)$, as in that setting, the expected value rises to 0.01.

Concretely, this means that when using $U(0,0.01)$ as the probability to catch an animal, the number of times an animal is being caught is twice as high as when using the three other probability distribution. Therefore, we expect the estimations of $\hat{N}$ and $\sigma_{\hat{N}}$ to be better when using $p \sim U(0,0.01)$ as opposed to the other distributions.

Trivially, as the expected behavior of the three other distributions are identical, we expect the estimations of

³https://en.wikipedia.org/wiki/Beta_distribution

$$p_i \sim U(0, 0.01)$$

		$N = 500$	$N = 1000$	$N = 5000$
$\hat{N}$		590.45±311	1096.62±286	5152.14±583
Error (%)		18.09	9.66	3.04
$\hat{\sigma}_{\hat{N}}$		239.58±258	284.17±116	566.36±98
C.I.	$\alpha = 0.05$	[299.68±85; 1322.77±1298]	[687.34±137; 1842.05±609]	[4179.35±423; 6414.78±804]
	$\alpha = 0.01$	[250.36±59; 1740.30±2079]	[601.62±110; 2185.30±773]	[3922.46±383; 6884.52±890]
Coverage (%)	$\alpha = 0.05$	94.50±22	95.70±20	94.70±22
	$\alpha = 0.01$	99.40±7	99.20±8	99.20±8
Bias ($\approx$)		8 181	9 334	23 147
Variance ($\approx$)		124 464	94 211	330 364
MSE ($\approx$)		132 646	103 546	353 512

Table 1: Result from 1000 MC simulations for estimating $\hat{N} \mid p_i \sim U(0, 0.01)$. For each value, we report on the mean (black) and standard deviation (gray). We omit the digits after the decimal point for the standard deviation for brevity. The Error (%) is the absolute difference between the estimated and the true value of N , divided by the true value of N .

$\hat{N}$ and $\sigma_{\hat{N}}$ to be very identical up to the effect of the randomness introduced by the Monte-Carlo simulations.

Observations. Let us first look at Table 1 which contains the result when using $p \sim U(0, 0.01)$.

First, it is very clear that increasing the true population size N decreases the relative error of the estimator. Furthermore, the estimated population size $\hat{N}$ is close to the true value of N for all three settings of our simulation with a deviation of approximately 90–150 animals.

Interestingly, although the absolute error of the population estimation increases with the true number of population, the relative error of the estimated population size to the true population size decreases with increasing true population size N . Whereas for a small population size ($N = 500$), the relative error is approximately 18%, for larger population sizes ($N = 5000$), the relative error drops to only 3%.

The estimated standard deviation ($\hat{\sigma}_{\hat{N}} = \sqrt{\hat{\sigma}_{\hat{N}}^2}$) from the given formula perfectly matches the standard deviation computed by the Monte-Carlo simulation (refer to the gray value next to the $\hat{N}$ estimate). Nevertheless, it is important to denote that the standard deviation of the estimated variance is very large (e.g., $\sigma_{\hat{\sigma}_{\hat{N}}} = \pm 258$ for $\hat{\sigma}_{\hat{N}} = 239.58$), hence, the estimated variance can deviate significantly across different runs. Therefore, the estimated variance is only trustworthy for large population sizes.

It is worth mentioning that although the standard deviation increases with increased population size N , the variance of the estimated standard deviation of the estimation of the population size decreases the larger N becomes.

We compute the coverage (percentage of times the true population size is contained in the confidence interval) for different values of $\alpha \in \{0.05, 0.01\}$ and observe that, regardless of the true population size N , the average coverage matches the expected coverage of the specified confidence intervals.

Interestingly, the standard deviation of the computed coverage from the Monte-Carlo simulation remains stable across all population sizes; We have a standard deviation of ± 22 percentage points for $\alpha = 0.05$ and ± 8 percentage points for $\alpha = 0.01$.

$$p_i \sim U(0, 0.02)$$

		$N = 500$	$N = 1000$	$N = 5000$
$\hat{N}$		527.70±98	1035.16±134	5088.17±284
Error (%)		5.54	3.52	1.76
$\hat{\sigma}_{\hat{N}}$		97.71±29	133.04±27	288.03±25
C.I.	$\alpha = 0.05$	[381.03±57; 774.58±172]	[818.85±91; 1347.52±198]	[4568.45±240; 5700.59±337]
	$\alpha = 0.01$	[348.62±49; 882.73±206]	[765.73±82; 1471.85±225]	[4421.98±228; 5914.73±356]
Coverage (%)	$\alpha = 0.05$	95.00±21	95.30±21	94.60±22
	$\alpha = 0.01$	99.40±7	98.80±10	98.70±11
Bias ($\approx$)		767	1 236	7 774
Variance ($\approx$)		10 398	18 460	83 612
MSE ($\approx$)		11 166	19 696	91 387

Table 2: Result from 1000 MC simulations for estimating $\hat{N} \mid p_i \sim U(0, 0.02)$. For each value, we report on the mean (black) and standard deviation (gray). We omit the digits after the decimal point for the standard deviation for brevity. The Error (%) is the absolute difference between the estimated and the true value of N , divided by the true value of N .

Finally, and most interestingly, we also compute the Bias, Variance, and the Mean Squared Error (MSE) for each context of our simulations. In Table 1, we observe that the bias only increases slightly from $N = 500 \rightarrow N = 1000$ (from $90^2 \rightarrow 96^2$) but increases drastically for $N = 1000 \rightarrow N = 5000$, where the bias increases by 2.5 times.

Most interestingly, the variance does not follow a simple pattern. While increasing $N = 500 \rightarrow N = 1000$ drops the variance by 23.7% (around 30 000 units of absolute difference), when increasing the true population size from $N = 1000 \rightarrow N = 5000$, we observe an increase of the variance of the estimation of the population size 350%.

Finally, because the variance is around 1 order of magnitude larger than the bias term, the MSE follows the general trend described in the variance.

Comparison to other Probability Distributions. While we went into details on the results from Table 1, these results result from $p \sim U(0, 0.01)$ and may therefore deviate from the results for the other probability distributions.

First, as hypothesized previously, the results between

- $p \sim U(0, 0.01)$ - See Table 1
- $p \sim 0.02 \cdot \text{Beta}(1, 3)$ - See Table 7
- $p \sim 0.01 \cdot \text{Beta}(2, 2)$ - See Table 8

are extremely similar. In fact, not only are the returned values all within the standard deviation of the results from each other — regardless of the true population size or the metric. Naturally, as each distribution has a different probability density function, and that Monte-Carlo simulation inherently introduces some noise, the results slightly differs between Table 1, Table 7, and Table 8.

However, when we look at Table 2, which contains the results for $p \sim U(0, 0.02)$, which has a higher expected value (see Equation 22), we observe that, indeed, the values differ (significantly) from the results of the other distributions.

First, the deviation of the estimated to the true population size is significantly lower for $p \sim U(0, 0.02)$ compared to the other distributions. The relative error is the perfect metric to quantify the change in deviation, as it decreases from 18% for $p \sim U(0, 0.01)$ to 5.54% for $p \sim U(0, 0.02)$ when $N = 500$, which is a decrease of 70%. Increasing the true population size to larger values has a similar effect. For example, for $N = 5000$, the relative error drops from 3% to 1.76%.

We remark that the estimated standard deviation and the true standard deviation of the estimated population size are very similar for $p \sim U(0, 0.02)$ too, with a lower standard deviation compared to the other distributions.

Similarly, the coverage remains stable across all population sizes, demonstrating that the confidence intervals are well-calibrated. Note that the standard deviation of the coverage is the same as for the other distributions, indicating that the std. of the coverage is potentially independent of the probability distribution used.

Finally, the MSE follows a different pattern as computed for the other distributions. While the bias increases with the same pattern but lower values as the other distributions, the variance also increases linearly with the size of the true population size. Therefore, the MSE roughly follows a linear increase with the size of the true population size.

However, the increase in the quality of the estimators is significant, as the MSE for $N = 5000$ drops from around 350 000 units to around 90 000 units, which is a decrease of around 75%.

Discussion — MSE. One of the most surprising observations is the change of the MSE w.r.t. the true population size. The results from Table 1 and Table 8 show that the MSE (and variance) follows a v shape when increasing the true population size, where $N = 1000$ has the lowest MSE when comparing to $N = 500$ and $N = 5000$.

We hypothesize that this phenomenon is simply due to the fact that the number of steps ($T = 40$) is too low to obtain accurate estimations when using probability distributions that have a low $\mathbb{E}[X_i^t = 1]$. We hypothesize that doubling the number of time steps should give us similar results as the results from Table 2 with $p \sim U(0, 0.02)$.

This hypothesis is supported by the results in Table 2 where the MSE follows a linear increase w.r.t. N .

To verify our hypothesis, we run the simulation using $p \sim U(0, 0.01)$ for twice as long, namely $T = 80$ instead of 40 time steps. We report on the results in Table 9 in the Appendix. In short, we observe that the results from Table 9 matches the results obtained in Table 2, therefore validating our hypothesis.

Discussion — Estimated Population Size. Interestingly, the estimated population size reported by our Monte-Carlo simulations always over-estimate the true population size. We want to understand this better, and understand whether this is a systematic property of the given estimator.

First, we hypothesize that most of the variance is due to the estimation of $\hat{f}_0$. As the given formula for estimating $\hat{f}_0$ is dependent on whether $f_2 \neq 0$ or $f_2 = 0$, we aim to first understand how similar both

formulas are. To this end, we plot in Figure 4 the estimated $\hat{N}$ over 100 time steps for a single run of the Monte-Carlo simulation with true population size $N = 1000$ and $p \sim U(0, 0.01)$.

In Figure 4, we clearly observe that, whenever $f_2 = 0 \rightarrow f_2 > 0$, the estimation of $\hat{f}_0$ is not continuous in time and drastically drops. Interestingly, as soon as $f_2 > 0$, the estimator of $\hat{N}$ seems to approach the true population size.

We also plot the true value for f_0 and the estimated value for $\hat{f}_0$ over the same number of time steps in Figure 4. We can clearly see that the estimated values for $\hat{f}_0$ approach the true value for f_0 over time.

We remark that the behavior observed in Figure 4 is not systematic, as the jump is very extreme. We found this example by chance when plotting the results for a single run. However, it provides us with great insights into the behavior of the estimators.

Derivation — Estimated Population Size.

In the remaining part of this section, we will investigate analytically whether the over-estimation of the population size is a systematic property of the given estimator.

Formally, we are interested in the following question

$$\Pr(\hat{N} > N) > \Pr(\hat{N} \leq N) . \quad (26)$$

To this end, we will assume that the estimator is unbiased, i.e., that $\mathbb{E}[\hat{N}] = N$. Note that Equation 26 is not in contradiction with the unbiasedness of the estimator, as the expected value is the value towards which the estimator converges to in the limit of the number of samples. However, for finite samples, even if the estimator has a theoretical mean of N , if the distribution of the estimated population size is skewed, it is possible that Equation 26 holds.

Before we start, let us first formally define some notation and definitions of f_i that allows us to derive the probability for $\hat{N} > N$.

Let us denote an animal as X_i with $i \in \{1, \dots, N\}$ and $X_i^t \in \{0, 1\}$ denoting whether the animal i has been caught at time step t . Therefore, we can define

$$X_i = \sum_{t=1}^T X_i^t ,$$

i.e., the total number of times the animal has been caught after T time steps.

Logically, we have that $X_i^t \sim \text{Bernoulli}(p)$ with p being the probability of being caught at time step t . In our setting, p follows a specified probability distribution.

We have that X_i^t are i.i.d. w.r.t. the time steps t and the animals i . These are important assumptions. From this, it follows that $X_i \sim \text{Binomial}(T, p)$ as we have a sum of T independent Bernoulli variables.

We can now formally define f_k as follows using the following set

$$S_k = \{X_i : i \in \{1, \dots, N\} \mid X_i = k\}$$

which contains all the animals that have been caught exactly k times within the T time steps. Therefore, f_k can be defined as follows:

$$f_k = |S_k|.$$

Because f_k is a random variable, we can define the probability distribution of f_k w.r.t. the number of time steps t as binomial distribution:

$$\Pr(f_k = n \mid T = t) = \text{Binomial}(N, n, \Pr(X_i = k \mid T = t)) \quad n \in \{0, 1, \dots, N\}$$

Therefore, we need to compute the probability of an animal being caught exactly k times within the T time steps. We have

$$\Pr(X_i = k \mid T = t') = \text{Binomial}(t', k, \Pr(X_i^t = 1))$$

where $\Pr(X_i^t = 1)$ is the probability of an animal being caught at time step t . Because X_i^t is a binary variable, its expected value is its probability, and we can therefore re-use the expected values for the different given probabilities that we computed earlier (see Equation 21, 22, 24, 25).

We can now compute the expected value for each f_k :

$$\begin{aligned} \mathbb{E}[f_k \mid T = t] &= \sum_{n=0}^N n \cdot \Pr(f_k = n \mid T = t) \\ &= \sum_{n=1}^N n \cdot \text{Binomial}(N, n, \Pr(X_i = k \mid T = t)) \\ &= \sum_{n=1}^N n \cdot \text{Binomial}(N, n, \text{Binomial}(t, k, p)) \\ &= \sum_{n=1}^N n \cdot \text{Binomial}\left(N, n, \binom{t}{k} p^k (1-p)^{t-k}\right) \\ &= \mathbb{E}\left[\text{Binomial}\left(N, n, \binom{t}{k} p^k (1-p)^{t-k}\right)\right] \\ &= N \cdot \binom{t}{k} p^k (1-p)^{t-k} \end{aligned}$$

	Expected Value for n		
N	500	1000	5000
$U(0, 0.01)$	90.83	181.67	908.39
$U(0, 0.02)$	165.51	331.02	1655.14

Table 3: Expected value for n for $T = 40$ and $N \in \{500, 1000, 5000\}$ for the different probability distributions.

Therefore, we can now compute the expected value for n :

$$\begin{aligned}
\mathbb{E}[n] &= \mathbb{E}\left[\sum_{k=1}^T f_k\right] \\
&= \sum_{k=1}^T \mathbb{E}[f_k] \\
&= \sum_{k=1}^T \mathbb{E}[f_k \mid T = T] \\
&= \sum_{k=1}^T N \cdot \binom{T}{k} p^k (1-p)^{T-k} \\
&= N \cdot \sum_{k=1}^T \binom{T}{k} p^k (1-p)^{T-k} \\
&= N \cdot \left[\left(\sum_{k=0}^T \binom{T}{k} p^k (1-p)^{T-k} \right) - \binom{T}{0} p^0 (1-p)^{T-0} \right] \\
&= N \cdot \left[1 - (1-p)^T \right] \\
&= N \cdot \left(1 - (1-p)^T \right) .
\end{aligned} \tag{27}$$

We report on the expected values for n with $T = 40$ for our four probability distributions in Table 3. Note that Equation 27 is equal to

$$\begin{aligned}
\mathbb{E}[n] &= N \cdot \left(1 - (1-p)^T \right) \\
&= N \cdot \left(1 - \binom{T}{0} p^0 (1-p)^T \right) \\
&= N - N \cdot \binom{T}{0} p^0 (1-p)^T \\
&= N - \mathbb{E}[f_0 \mid T = T]
\end{aligned}$$

which is exactly what we would expect.

Because $\hat{N} = n + \hat{f}_0$, we now compute the expected value for $\hat{f}_0$. As $\hat{f}_0$ is conditional on whether $f_2 \neq 0$ or $f_2 = 0$, we need to compute the probability of $f_2 = 0$ and $f_2 \neq 0$. Because we model f_k as a binomial

	$\Pr(f_2 = 0 \mid T = 40)$		
N	500	1000	5000
$p = 0.01$	1.3×10^{-12}	1.7×10^{-24}	1.6×10^{-119}
$p = 0.005$	2.96×10^{-4}	8.77×10^{-8}	5.19×10^{-36}

Table 4: Probability for $f_2 = 0$ for $T = 40$ and $N \in \{500, 1000, 5000\}$ for the different probability distributions.

	$\mathbb{E}[\hat{f}_0 \mid f_2 = 0, T = 40]$		
N	500	1000	5000
$p = 0.01$	9113.93	36492.25	913036.85
$p = 0.005$	3375.21	13514.37	338129.79

Table 5: Expected value for $\hat{f}_0$ conditioned on $f_2 = 0$ for $T = 40$ and $N \in \{500, 1000, 5000\}$ for different values of p .

distribution, we can easily compute the probability of $\Pr(f_2 = 0)$. We have

$$\begin{aligned} \Pr(f_2 = 0 \mid T = T) &= \text{Binomial}(N, 0, \Pr(X_i = 2 \mid T = T)) \\ &= (1 - \Pr(X_i = 2 \mid T = T))^N \\ &= (1 - \text{Binomial}(T, 2, p))^N. \end{aligned}$$

We report on the probabilities for our four probability distributions for p in Table 4. We also plot the probability changing over time in Figure 5.

Before investigating the case where $f_2 \neq 0$, let us first investigate the bias of $\hat{f}_0 \mid f_2 = 0$. We have

$$\begin{aligned} \mathbb{E}[\hat{f}_0 \mid f_2 = 0, T = t] &= \mathbb{E}\left[\frac{f_1^2 - f_1}{2}\right] \\ &= \sum_{n=0}^N \frac{n^2 - n}{2} \cdot \Pr(f_1 = n \mid T = t) \\ &= \sum_{n=0}^N \frac{n^2 - n}{2} \cdot \text{Binomial}(N, n, \Pr(X_i = 1 \mid T = t)) \\ &= \sum_{n=0}^N \frac{n^2 - n}{2} \cdot \text{Binomial}(N, n, \text{Binomial}(t, 1, p)) \\ &= \sum_{n=0}^N \frac{n^2 - n}{2} \cdot \text{Binomial}(N, n, \binom{t}{1} p^1 (1-p)^{t-1}) \end{aligned}$$

which we cannot simplify further (at least such that it gives us more insights). We use python to compute the expected value for $\hat{f}_0 \mid f_2 = 0, T = 40$ for our four probability distributions and report on the results in Table 5. While the expected values for $\hat{f}_0 \mid f_2 = 0$ are very large, it is easy to see that when multiplied with the probability of $f_2 = 0$, we would get values that are almost zero. The interested reader can find a plot describing the rescaled expected value in Figure 6 in the Appendix.

Let us now investigate the bias of $\hat{f}_0$ when $f_2 \neq 0$. As $\hat{f}_0$ is defined as a function of f_1 and f_2 , and that both of them are dependent,⁴ we have to model their probability jointly for a fully analytical derivation. While it is very tempting to add a independence assumption on the distributions of f_1 and f_2 , their correlation is too strong to be ignored, and the obtained results are not meaningful.⁵ We show the correlations for our probability distributions in Table 11 in the Appendix.

The only way to compute $\mathbb{E}[\hat{f}_0 \mid f_2 > 0]$ is by marginalizing over all possible values for all f_i with $f_i \notin \{1, 2\}$, i.e.,

$$\begin{aligned} \mathbb{E}[\hat{f}_0 \mid f_2 > 0] &= \mathbb{E}\left[\frac{f_1^2}{2f_2} \mid f_2 > 0\right] \\ &= \sum_{n=0}^{N-1} \sum_{m=1}^{N-n} \frac{n^2}{2m} \cdot \Pr(f_1 = n, f_2 = m \mid T = t) \\ &= \sum_{n=0}^{N-1} \sum_{m=1}^{N-n} \frac{n^2}{2m} \cdot \left(\sum_{k_3=0}^{N-n-m} \cdots \sum_{k_N=0}^{N-n-m-\sum_{i=3}^N k_i} \Pr(f_1 = n, f_2 = m, f_3 = k_3, \dots, f_N = k_N \mid T = t) \right) \end{aligned} \quad (28)$$

which is computationally infeasible. Furthermore, we cannot simplify Equation 28, as every f_i is dependent on each f_j with $i \neq j$. Note that especially for our simulation size, using $N = 5000$ would result in a value proportional to 2^{5000} , which is a number with 1506 digits.

Therefore, without further assumptions, we are unsure whether we can show that N is unbiased — let alone skewed — in a fully analytical way.

Nevertheless, as we can model individual f_i using a binomial distribution, we can model the skewness of both f_1 and f_2 analytically. This might give us some insights into whether Equation 26 holds.

Skewness

Let us now investigate the skewness of $f_i \sim \text{Binomial}(T, p_i)$. It is a well known fact that the skewness of a binomial distribution is given by

$$\text{Skewness}(f_i) = \frac{(1-p) - p}{\sqrt{N \cdot p \cdot (1-p)}}.$$

⁴We attach in the Appendix (See Section .1) a proof of the dependency of f_1 and f_2 generated by ChatGPT. As the proof is not critical for the continuation of this derivation, we add it to the Appendix for the interested reader. Note that while we have deleted the exact output of ChatGPT, the idea of this proof is still to be credited to ChatGPT.

⁵This was our first attempt. However, our formula estimated $\mathbb{E}[\hat{f}_0 \mid f_2 > 0, T = 40]$ to be almost zero for all probability distributions and sample sizes.

	Skewness(f_1)			Skewness(f_2)		
N	500	1000	5000	500	1000	5000
$p = 0.01$	0.0463	0.0327	0.0146	0.17799	0.12585	0.05628
$p = 0.005$	0.0809	0.0572	0.0256	0.34368	0.24302	0.10868

Table 6: Skewness for f_1 and f_2 for $T = 40$ when modeled as a binomial distribution.

For f_1 , we have that

$$\begin{aligned}
\text{Skewness}(f_1) &= \frac{(1 - \Pr(X_i = 1 \mid T = T)) - \Pr(X_i = 1 \mid T = T)}{\sqrt{N \cdot \Pr(X_i = 1 \mid T = T) \cdot (1 - \Pr(X_i = 1 \mid T = T))}} \\
&= \frac{\left(1 - \binom{T}{1} p^1 (1-p)^{T-1}\right) - \binom{T}{1} p^1 (1-p)^{T-1}}{\sqrt{N \cdot \left(1 - \binom{T}{1} p^1 (1-p)^{T-1}\right) \cdot \binom{T}{1} p^1 (1-p)^{T-1}}} \\
&= \frac{1 - 2 \cdot \binom{T}{1} p^1 (1-p)^{T-1}}{\sqrt{N \cdot \left(1 - \binom{T}{1} p^1 (1-p)^{T-1}\right) \cdot \binom{T}{1} p^1 (1-p)^{T-1}}}
\end{aligned}$$

which we will compute using python. This results in the results shown in Table 6.

Recall that a positive value for skewness indicates that the distribution has a median and mode on the left of the mean, and therefore the majority of the values are on the left, and the distribution is tailed to the right. A negative value has the same interpretation, but flipped. We added a visualization of the interpretation of the value of skewness in Figure 7 in the Appendix.

From Table 6, it is easy to see that all values are positive. Therefore, we know that f_1 and f_2 have the majority of their values to the left of the mean, i.e., have a lower value than the mean. Naturally, the more samples we have, the less skewed the distribution is, this is also reflected in the table.

We are unable to compute the skewness of n , which follows the same argument as above, namely that all the f_i are dependent on each other. While all the f_i can be modeled using a multinomial distribution, a closed form solution to the skewness of a multinomial distribution is unknown to us, we couldn't find any literature on the topic. Therefore, we assume that n is either not skewed, or very lightly skewed in the opposite direction of $\hat{f}_0$. Otherwise, our argument does not hold.

However, and this is very interesting, we can say something about the skewness of $\hat{f}_0$. When looking at the skewness of $\hat{f}_0$ in Table 6, we see that f_1 is less skewed than f_2 . In fact, f_2 is skewed around 4–5 times more than f_1 . This is a very important and interesting observation.

This shows that, for small and finite samples, we are more likely to sample values for f_1 and f_2 below the mean than above. The inequality in the skewness from Table 6 indicates that, f_2 is sampled more below the expected value of f_2 than compared to f_1 .

As we divide the less undersampled value by the more undersampled value, we argue that $\hat{f}_0$ is negatively skewed. This follows from the argument that, assuming that $\mathbb{E}[\hat{f}_0] = f_0$ holds, we would $\frac{f_1}{f_2} \approx 5$, which would mean that we overestimate f_0 .⁶ As we are squaring the f_1 value, and only increasing the value of f_2 linearly

⁶We have not used skewness in any work before. Therefore, we are unsure if this is correct. However, that this ration is

in $\hat{f}_0$, we believe that this reduces the extremeness of the skewness of $\hat{f}_0$ significantly.

Therefore, assuming that $\mathbb{E}[\hat{f}_0] = f_0$ holds, we argue that Equation 26 holds. How to prove this formally is unknown to us; We believe that this statement is not true in general — as in the limit, the skewness, variance, and bias becomes zero — but we argue that especially for small finite samples, the statement holds.

We tried to get some help from ChatGPT, as we hope that it can uncover some theorems unknown to us. While it gave us a negative answer on whether the statement holds, it said that if our distribution is log-concave, then $P(X < E[X]) \geq \frac{1}{2}$ holds for some random variable $X \sim G$ with G being a log-concave positively skewed distribution. As the Binomial distribution is log-concave,⁷ we *could* argue that because we hypothesize that $\frac{f_1^2}{2f_2}$ is negatively skewed, we have $P(\hat{f}_0 > f_0) \geq \frac{1}{2} \Rightarrow P(\hat{f}_0 > f_0) > P(\hat{f}_0 \leq f_0)$. But as we are unsure about the correctness of our argument, we will only mention it here.

larger than 1 is what we are trying to argue.

⁷See https://en.wikipedia.org/wiki/Logarithmically_concave_function. We are unsure if this is correct for all cases, especially ours.

Statement on the Useage of Gen. AI

Unless stated otherwise, all code, text, and math derivations have been entirely thought and written by me with no external help — both for gen. AI and wolframalpha and co.

We used Claude 4.5 Opus for generating the plots for Figure 2 as the plot is not a core of the assignment, and only aids in understanding our own results. Note that we (heavily) modified the code generated by Claude.

For question 2 of the assignment, we used ChatGPT 5.2 to help us with some of the math of the derivations. The end product is 90% my own, indepent work. Most of the A.I. being used for question 2 got scrapped as it did not add any value to the report. The most notable scrapped work was finding some result to $\mathbb{E} \left[\frac{f_1^2}{2f_2} \right]$, but as both are dependent, only very limited results were found by ChatGPT.⁸

Additional Tables

$$p_i \sim 0.02 \cdot \text{Beta}(1, 3)$$

		$N = 500$	$N = 1000$	$N = 5000$
$\hat{N}$		572.21±249	1102.16±327	5165.82±579
Error (%)		14.44	10.22	3.32
$\hat{\sigma}_{\hat{N}}$		223.18±168	288.20±146	568.22±96
C.I.	$\alpha = 0.05$	[296.06±80; 1241.40±822]	[688.71±147; 1861.17±749]	[4189.85±421; 6432.61±798]
	$\alpha = 0.01$	[248.36±58; 1610.48±1203]	[602.47±115; 2212.82±975]	[3932.12±381; 6903.90±882]
Coverage (%)	$\alpha = 0.05$	94.60±22	94.00±23	94.60±22
	$\alpha = 0.01$	98.90±10	98.80±10	98.90±10
Bias ($\approx$)		5 214	10 436	27 495
Variance ($\approx$)		78 120	104 540	332 210
MSE ($\approx$)		83 334	114 977	359 705

Table 7: Result from 1000 MC simulations for estimating $\hat{N}$. For each value, we report on the mean (black) and standard deviation (gray). We omit the digits after the decimal point for the standard deviation for brevity. The Error (%) is the absolute difference between the estimated and the true value of N , divided by the true value of N .

.1 Proof of the dependency of f_1 and f_2

Custom Note:

The idea of this proof is to be credited to ChatGPT.

The current proof is just our own interpretation of the proof generated by ChatGPT, which we have deleted in the mean time.

We can model the probability distribution of all f_k jointly as a multinomial distribution.

⁸This may also be due to my bad prompting.

$$p_i \sim 0.01 \cdot \text{Beta}(2, 2)$$

		$N = 500$	$N = 1000$	$N = 5000$
$\hat{N}$		582.90±314	1093.58±294	5128.88±565
Error (%)		16.58	9.36	2.58
$\hat{\sigma}_{\hat{N}}$		235.15±275	281.72±121	561.35±93
C.I.	$\alpha = 0.05$	[297.63±84; 1303.01±1391]	[687.38±139; 1831.89±634]	[4164.29±411; 6379.84±776]
	$\alpha = 0.01$	[249.11±59; 1716.36±2272]	[602.16±110; 2171.47±808]	[3909.42±373; 6844.99±858]
Coverage (%)	$\alpha = 0.05$	95.30±21	95.50±20	95.40±20
	$\alpha = 0.01$	99.10±9	99.10±9	98.80±10
Bias ($\approx$)		6 871	8 757	16 608
Variance ($\approx$)		131 042	94 106	323 905
MSE ($\approx$)		137 913	102 864	340 514

Table 8: Result from 1000 MC simulations for estimating $\hat{N}$. For each value, we report on the mean (black) and standard deviation (gray). We omit the digits after the decimal point for the standard deviation for brevity. The Error (%) is the absolute difference between the estimated and the true value of N , divided by the true value of N .

$$p_i \sim U(0, 0.01)$$

		$N = 500$	$N = 1000$	$N = 5000$
$\hat{N}$		517.55±100	1022.24±129	5057.81±290
Error (%)		3.51	2.22	1.16
$\hat{\sigma}_{\hat{N}}$		95.08±31	130.52±27	285.44±26
C.I.	$\alpha = 0.05$	[374.85±57; 757.81±181]	[809.99±88; 1328.62±193]	[4542.75±244; 5664.70±345]
	$\alpha = 0.01$	[343.31±49; 863.09±218]	[757.85±78; 1450.54±219]	[4397.59±232; 5876.88±364]
Coverage (%)	$\alpha = 0.05$	95.00±21	95.20±21	94.60±22
	$\alpha = 0.01$	98.80±10	99.60±6	98.40±12
Bias ($\approx$)		307	494	3 342
Variance ($\approx$)		10 013	17 769	82 162
MSE ($\approx$)		10 321	18 263	85 504

Table 9: Same configuration as for Table 1, but we run the simulation for twice as long ($T = 80$).

In statistics, we can define two variables as independent if their covariance is zero.

From the multinomial distribution, we know that the covariance between f_i and f_j is given by

$$\text{Cov}(f_i, f_j) = -Np_i p_j .$$

Therefore, we have that

$$\begin{aligned}
\text{Cov}(f_1, f_2) &= -Np_1 p_2 \\
&= -N \cdot \Pr(X_i = 1 \mid T = t) \cdot \Pr(X_i = 2 \mid T = t) \\
&= -N \cdot \text{Binomial}(T, 1, p) \cdot \text{Binomial}(T, 2, p)
\end{aligned}$$

Naturally, we see that the Covariance is not zero, but slightly negative, and therefore the two variables are

not independent.

For our four probability distributions, we have the following covariance values:

	Cov(f_1, f_2)		
N	500	1000	5000
$p = 0.01$	-7.20	-14.39	-71.95
$p = 0.005$	-1.33	-2.65	-13.26

Table 10: Covariance for f_1 and f_2 for $T = 40$ and $N \in \{500, 1000, 5000\}$ for different values of p .

From the multinomial distribution, we know that the variance of f_i is given by

$$V(f_i) = Np_i(1 - p_i) . \quad (29)$$

Therefore, we can use Equation 29 to compute the correlation between f_1 and f_2 :

	Corr(f_1, f_2)		
N	500	1000	5000
$p = 0.01$	-0.1443	-0.1443	-0.1443
$p = 0.005$	-0.0568	-0.0568	-0.0568

Table 11: Covariance for f_1 and f_2 for $T = 40$ and $N \in \{500, 1000, 5000\}$ for different values of p .

□

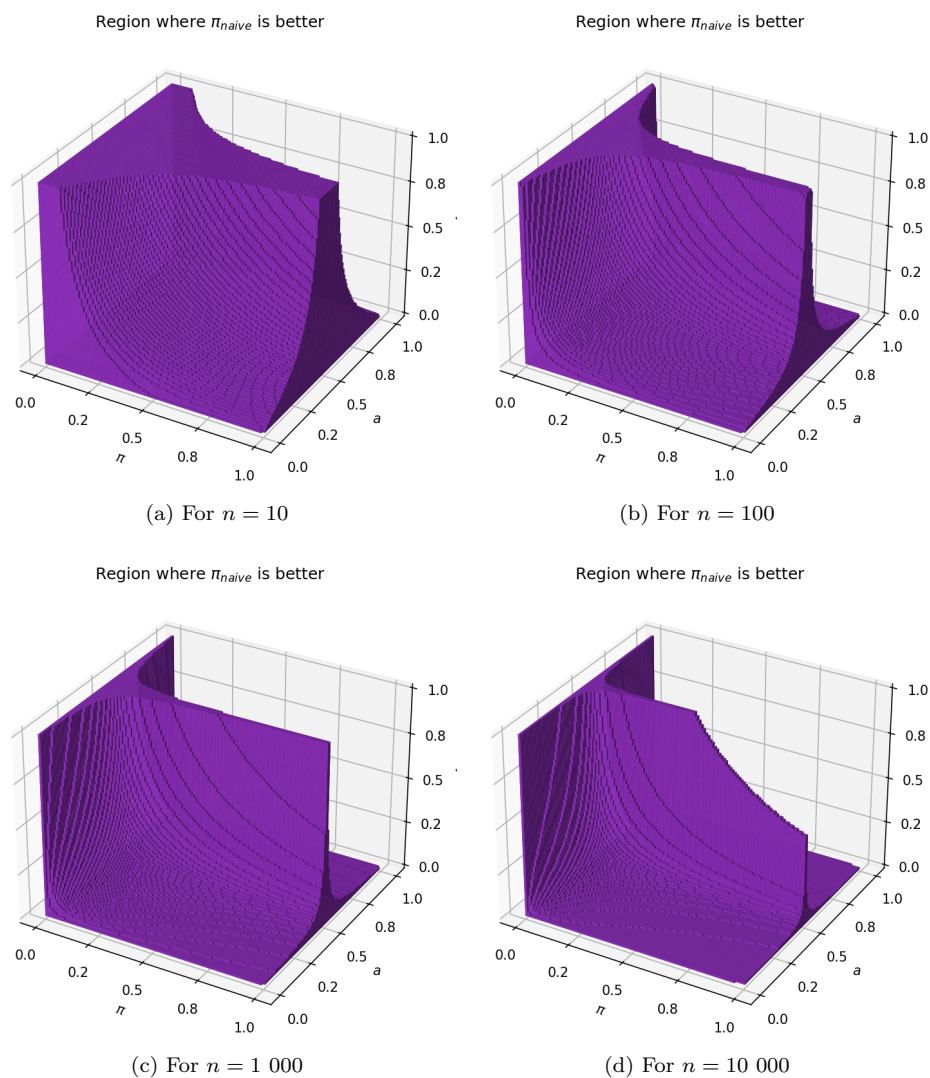


Figure 2: Plots of the parameter space of (π, a, ℓ) for various sample sizes n . The purple voxels indicate for which parameters the MSE of π_{naive} is lower than the MSE of π_{ML} . Note that plot is mirrored around the ℓ -axis at $a = 0.5$.

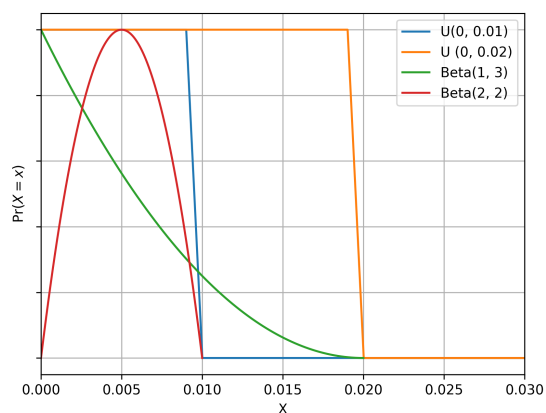


Figure 3: Probability density functions for the four different probability distributions. We normalize the probability density functions to have a maximum value of 1 for each distribution as we are interested in the relative differences between the different distributions. While the height of the pdfs are not to scale, their shape are.

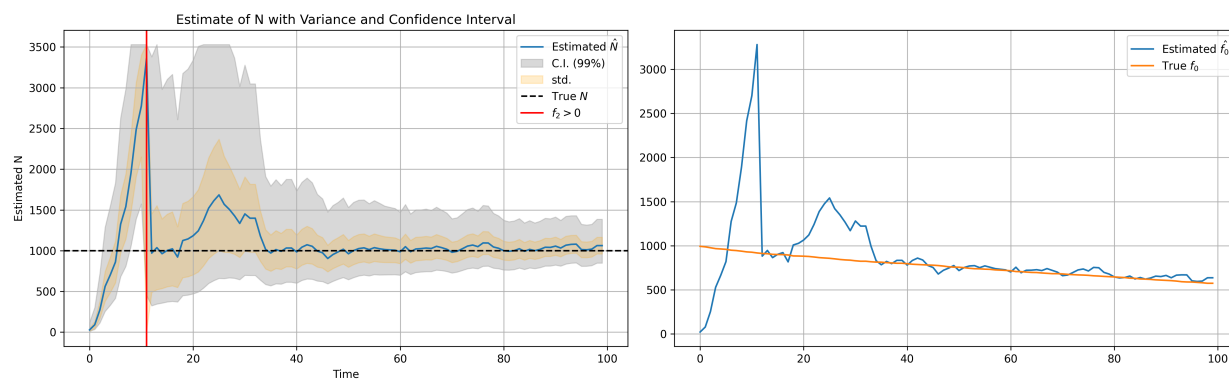


Figure 4: Single run of the Monte-Carlo simulation for estimating $\hat{N}$ with a true population size of $N = 1000$ and a probability distribution of $p_i \sim U(0, 0.01)$.

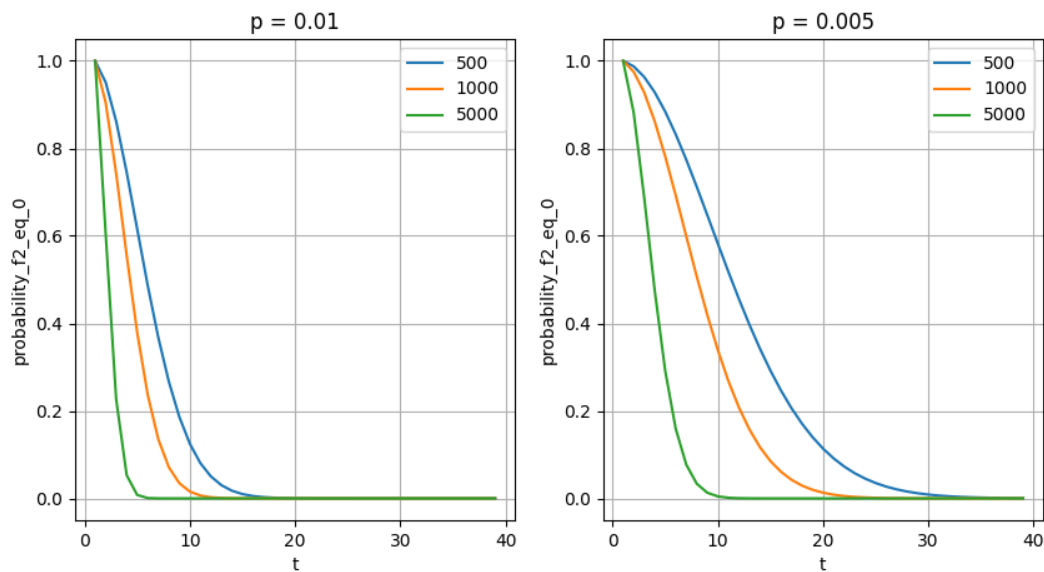


Figure 5: Probability for $f_2 = 0$ over 40 times steps for different population sizes. Note that

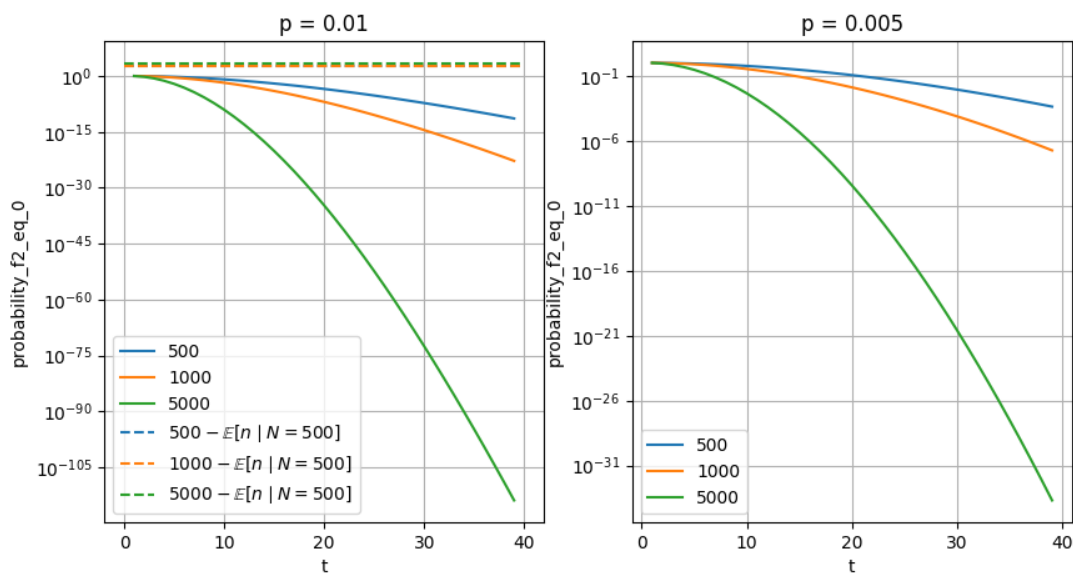


Figure 6: The expected value for $\hat{f}_0$ conditioned on $f_2 = 0$ over 40 times steps for different population sizes. We multiply the expected value by the probability of $f_2 = 0$ at each time step.

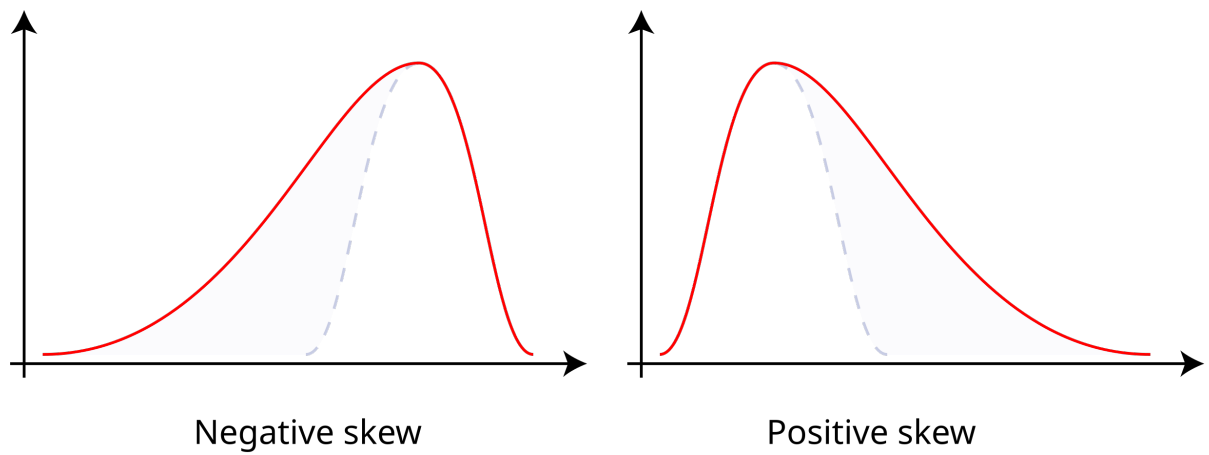


Figure 7: Interpretation of the skewness of a distribution.